

PolyU-CALLS 2025

**Developing and Evaluating STEM Vocabulary for Effective
Learning and Teaching:
*Approach to Word Selection and Survey Design***

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1. Introduction: Why STEM Vocabulary Matters

- **STEM:** Science, technology, engineering, and mathematics

- **Background:**
 - 1) **Mastering STEM-specific vocabulary** is essential for STEM undergraduate academic success (Yahya et al., 2022)
 - 2) In the context of establishing **Hong Kong's third medical school**, we conducted a **pilot study** focusing **medical vocabulary** for English for Specific Purposes (ESP).

- **The gap:** **Limited existing resources** address the **unique challenges** of STEM vocabulary acquisition

1. Introduction: Why STEM Vocabulary Matters

- **Unique challenges of STEM vocabulary acquisition:**
 1. Difficulty for Non-native English Speakers
 2. Complexity of Professional Terminology
 3. Needs of Interdisciplinary Vocabulary acquisition (Xu & Ouyang, 2022)
 4. *Lack of Effective Learning Resources and Strategies*
 5. Cognitive Load (Lobos et al., 2024)

- **Our goal:** Evaluate **key characteristics of STEM vocabulary** to **support effective learning & teaching**

2.1 Methodology: The Power of Affixes

- Affixes have **Prefixes** (Figure 1) & **Suffixes** (Figure 2)

super-	non-	fore-	
ir-	sub-	de-	pre-
under-	mis-	anti-	
over-	em-	inter-	

Figure 1

-able	-s	-less	
-ive	-ful	-ment	
-en	-y	-ity	-er
-est	-ed	-tion	

Figure 2

2.1 Methodology: The Power of Affixes

➤ Key Insight:

- **61 distinct affixes** found in **138 medical vocabulary** (e.g., *poly-*, *radio-*, *-ar*, *-logist* ...)

➤ Anchor Words:

- **21 high-frequency words with common affixes** (e.g., *micro-*, *-scopy*, *-ium* ...)
- **Rationale:**
 - ↑ the efficiency of the survey
 - control variables in each survey for better statistical analysis (help ↑ statistical significance)
 - verify the role of affixes in medical vocabulary learning → **Affixes simplify vocabulary learning patterns** (Namsaeng et al., 2024) (Yuan & Tang, 2023) (Saigh & Sonbul, 2018)

2.2 Methodology: Anchor Word Selection



- 1. Ask ChatGPT to **identify** all vocabularies' **affixes** (prefixes & suffixes)
- 2. Ask ChatGPT to **categorise** the affixes present in the vocabularies **in the lists that fit in the Excel format**
- 3. **Selected** the same affixes in words that appear **≥ 2 times** → **21 affix groups**
(also check again by myself)

poly-	Frequency	re-	Frequency	-ic	Frequency	-ation	Frequency
polymer	65553	recombination	7441	caloric	22377	hydration	57999
polymerase	14546	respiration	21345	diastolic	7818	immobilization	5526
polymorphism	6551	reflex	41584	genetic	306536	dilation	11614
				organic	603812	fertilization	19999
				parasitic	22960	hybridization	8486
				ultrasonic	23229	fermentation	58847
				phenotypic	5206	recombination	7441
				acrylic	98485	radiation	279592
				toxic	233035		
				microscopic	40046		
				anaerobic	21932		
				aerobic	42853		

Figure 3

2.2 Methodology: Anchor Word Selection



- **4. Searched the word frequencies** using the **iWeb** for each word in the affix group
- **5. Selected the highest-frequency words** from each affix group, creating a list of **21 anchor words**

poly-	Frequency	re-	Frequency	-ic	Frequency	-ation	Frequency
polymer	65553	recombination	7441	caloric	22377	hydration	57999
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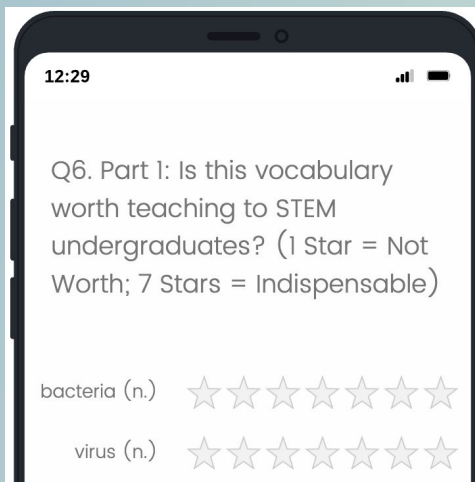
p.s. Remaining words be randomly divided into 9 groups, called **random words**

Figure 3

2.3 Methodology: The Design of the Survey

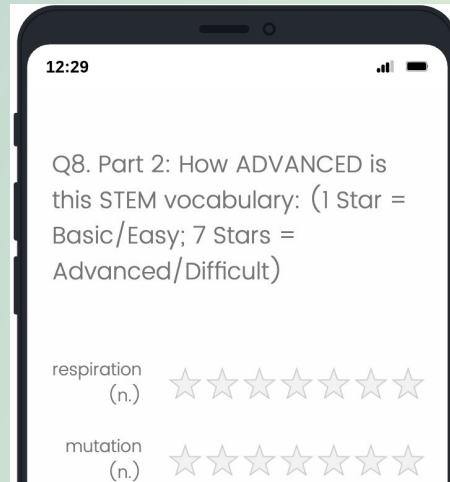
- **Participants (Raters):** Medical subject's Professors, Instructors, PhD students (TAs)
- **What They Do:** Rate words on **usefulness & difficulty** (7-point likert scale)
 - 21 anchor words + 13 random words per survey → $(21 + 13 \times 9 = \text{in total } 138 \text{ be rated})$
 - We made 9 different survey
 - Around 5 minutes to complete, and then rater will be rewarded a coffee voucher

(He & Godfroid, 2019)



**Usefulness
question**

Figure 4



**Difficulty
question**

Figure 5

2.3 Methodology: The Design of the Survey

➤ Structure of our survey's content:

Background Information

Rating Part 1:

Usefulness_Anchor words (21 words)

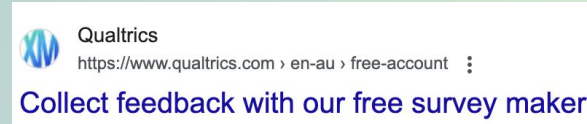
Usefulness_Random words (13 words)

Rating Part 2:

Difficulty_Anchor words (21 words)

Difficulty_Random words (13 words)

➤ (He & Godfroid, 2019)



To minimize confusion, I **randomized the word order** in the **Difficulty section**, as it has same words with the Usefulness section.

3. Implementation

Pilot: Build the Medical wordlist (briefly divide into 4 steps)

- **Step 1: Cross-reference:**
- **HKUST STEM Vocabulary website** & Medical vocabularies in **COCA (Corpus of Contemporary American English) Academic Wordlists**
- **Our wordlists unique part: adopt a morphology (affix-based) approach**

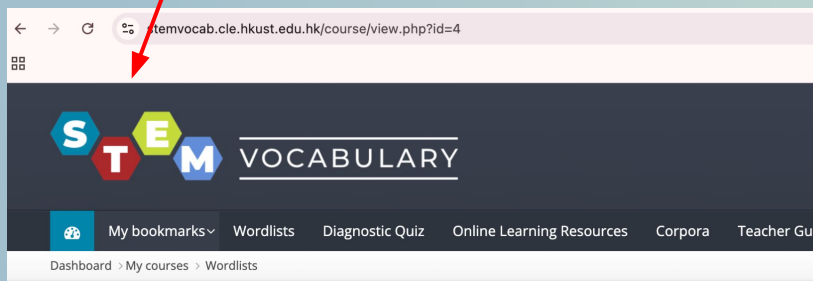


Figure 6

Mechanical and Aerospace Engineering Wordlist

Chemical and Biological Engineering Wordlist

Civil and Environmental Engineering Wordlist

Chemistry Wordlist

Figure 7

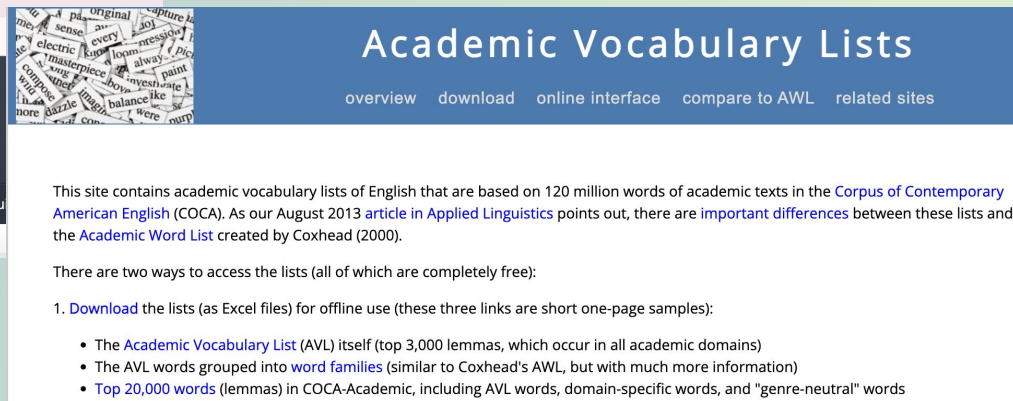


Figure 8

3. Implementation

Pilot: Build the Medical wordlist (briefly divide into 4 steps)

- **Step 2:** In this pilot, preliminary selected 138 relatively common **Medical vocabularies**
- **Step 3:** Categorize **Medical vocabulary** in groups by their **frequency & affixes** to create a focused experimental wordlist.
- **Step 4:** Find expert raters to rate



Figure 9

iWeb: a comprehensive corpus based on COCA-Academic and larger than it

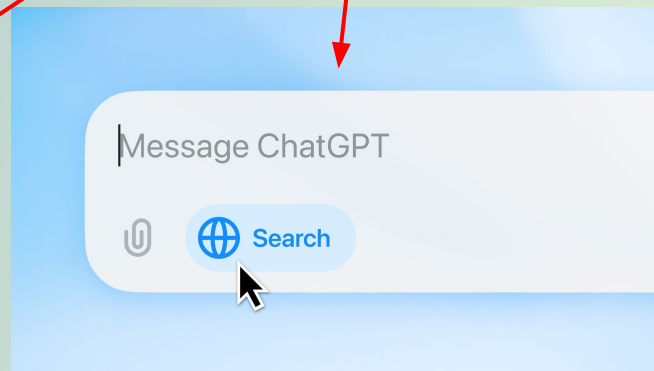


Figure 10

4. Results and Analysis

- In our wordlists:
 - (44 in 138 words) **32% of medical vocabularies** are **without affixes** (e.g., agar, shunt, resin, acid ...)
 - **Remaining 68% of medical vocabularies with affixes** → support our approach to affix-based vocabulary learning strategies.
- After collecting the surveys from the raters, conduct the **data analysis** on **expert subjective word difficulty and usefulness ratings** → help to create more effective learning materials

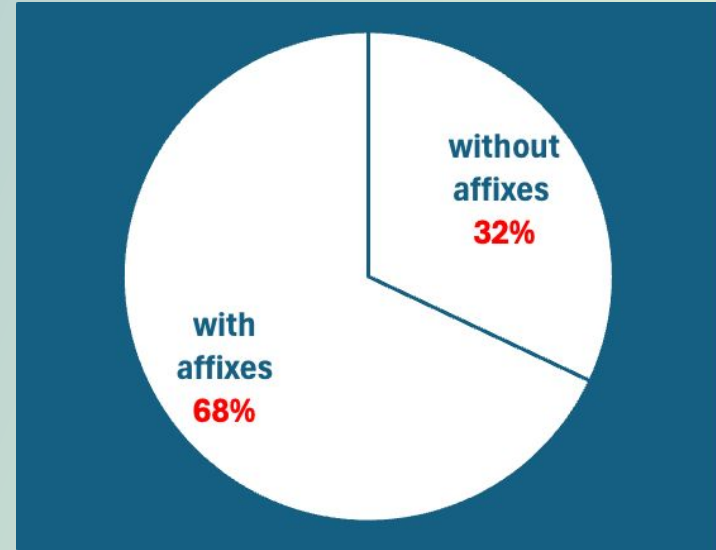


Figure 11

5. Summary and Next Steps

➤ Key Takeaways

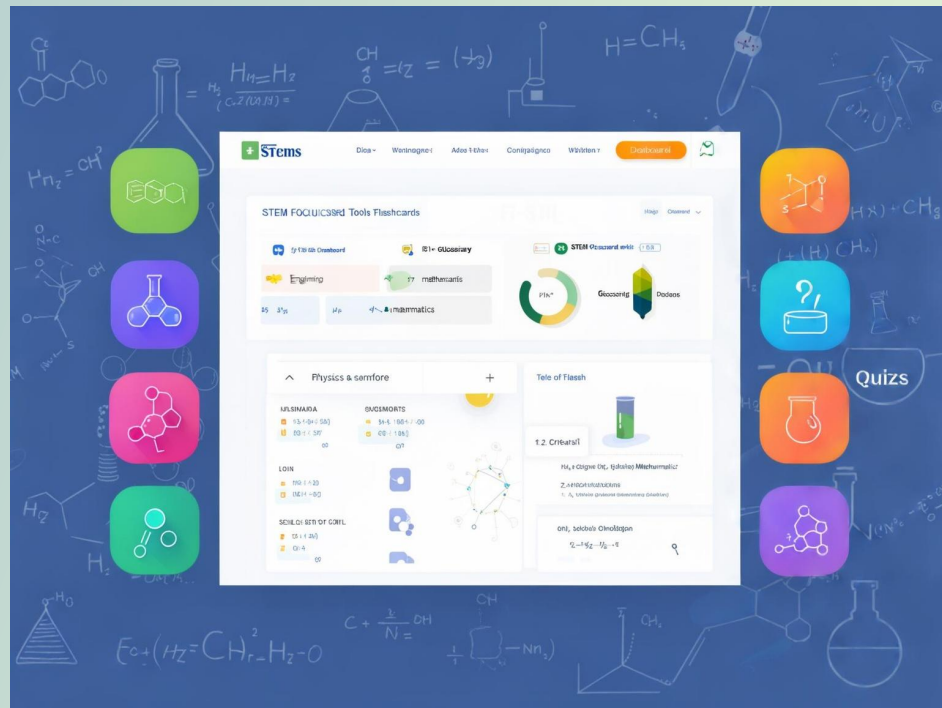
- Around 70% medical vocabulary **with affixes**.
- Learning and teaching STEM vocabulary needs **corpus-driven, affix-aware** solutions.
- Use a combination of **objective and subjective data**, to create more effective vocabulary learning materials.
- Base on STEM vocabulary categorization & selection, we need support from **specific experts**. E.g., PhD candidates (teaching assistants), ESL instructors, and professors (instructors)

5. Summary and Next Steps

➤ Next Steps

➤ **For Researchers:**

- 1) Expand this words categorization and rating method → **engineering, physics, mathematics, chemistry, etc. other STEM subjects**
- 2) Build a STEM words learning website → provide **subject-specific & affix-based** wordlists and learning approaches



5. Summary and Next Steps

➤ For Educators:

Develop **focused teaching materials**, such as digital flashcards, glossaries, and quizzes based on affix patterns

→ align with the contextual and morphological characteristics of the terms.

[illegible]

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7. Q&A

Thanks for listening!
Any feedback would be appreciated!

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